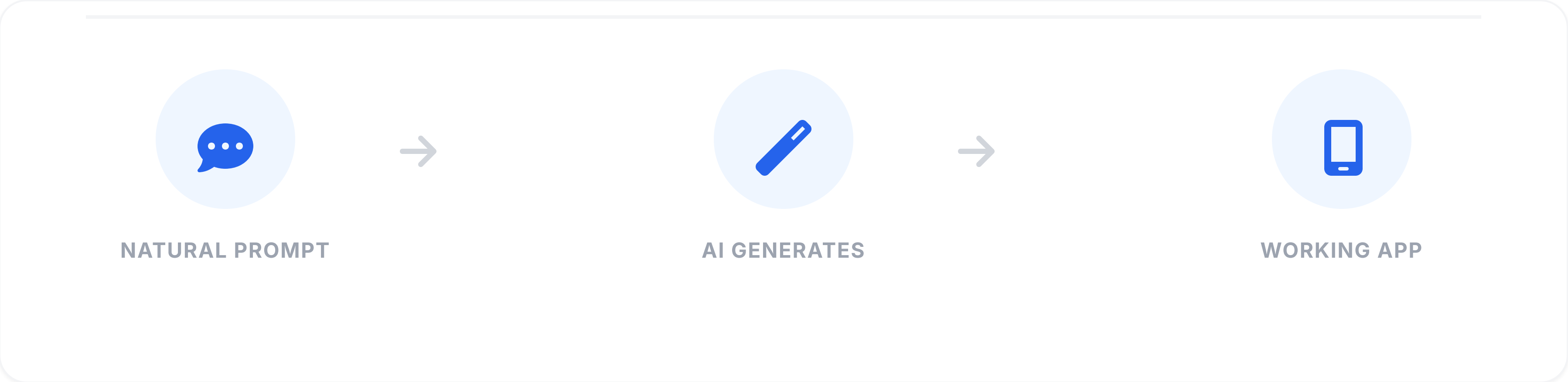
MODULE 4

Introduction to Vibe Coding

Build apps by describing what you want—then refine together with AI tools.



What You Will Learn

Vibe Coding Basics

Understand what vibe coding is and why it's a powerful new way to build digital solutions.

New vs. Traditional

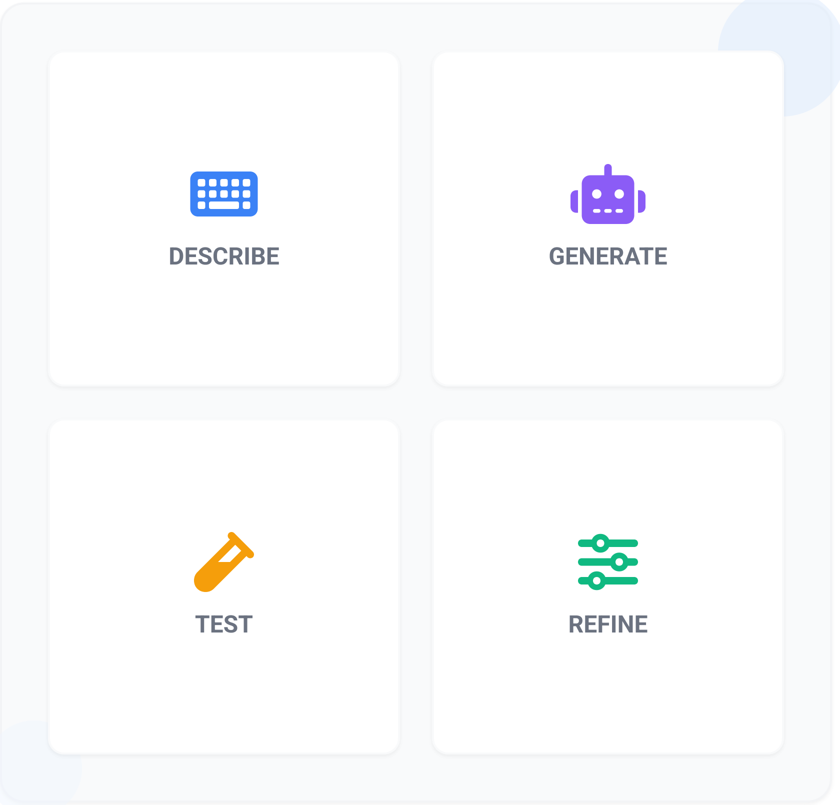
Learn how vibe coding differs from traditional software development methods.

The Core Cycle

Master the 4-step process: Describe → Generate → Test → Refine.

Prompting & Iteration

Discover how to write effective prompts and use iteration to speed up prototyping.



What is Vibe Coding?

> **DEFINITION**
>
> Vibe coding is the process of building digital solutions by **describing what you want** in natural language instead of writing code manually line by line.

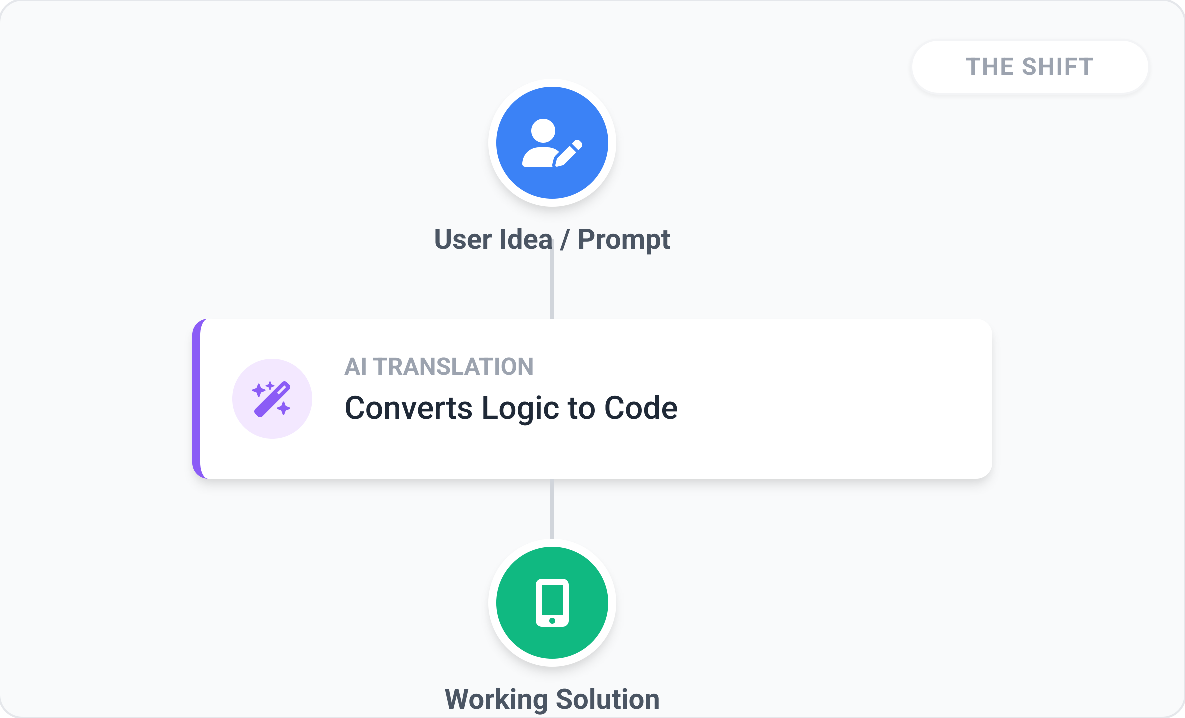
Why It Matters

- ✓ Move from idea to prototype quickly
- ✓ Build without deep programming expertise
- ✓ Focus on the outcome rather than the syntax

In Simple Terms

Instead of: "I need to program this from scratch..."

You say: "I want an app that does this, includes these features, and looks like this."



"Vibe coding changes the interface of creation from commands to conversation."

Traditional Development vs Vibe Coding

COMPARISON

Traditional Development

CODE-FIRST APPROACH

- **Developer-centric**, manual implementation
- Slower iteration & testing cycles
- High technical barrier (requires coding skills)
- → Plan → Design → Code → Test

THE OLD WAY

"Build this by writing all the code manually."

Vibe Coding

PROMPT-FIRST APPROACH

- ✓ **User-centric**, outcome-driven
- ✓ Rapid iteration via conversational prompts
- ✓ Lower technical barrier (accessible to more)
- → Describe → Generate → Test → Refine

THE NEW WAY

"Describe what you want, then refine the result."

COMPARISON

ITERATION SPEED	BARRIER TO ENTRY	IMPLEMENTATION
Slower → Rapid	High Technical → Accessible	Manual Coding → AI-Generated

Key Message: Vibe coding does not remove design thinking—it changes how humans communicate with the system.

MODULE 4 VIBE CODING CYCLE

How Vibe Coding Works

Vibe coding follows a simple iterative cycle: **Describe** → **Generate** → **Test** → **Refine**.

THE 4-STEP CYCLE

1. Describe

Explain purpose, users, features, & constraints in natural language.

2. Generate

AI drafts the interface, logic, workflow, and data structure.

3. Test

Review the prototype. Does it meet goals? Is it usable?

4. Refine

Adjust with new prompts. This loop is the key to quality.

EXAMPLE PROMPT

```
"Create an application that allows users to enter records, track changes over time, and display alerts when values
exceed a threshold."
```

Key Message: Vibe coding is not a one-shot process. It is an iterative conversation between the user and the AI tool.

Prompting in Vibe Coding

BEST PRACTICES

Weak Prompt

VAGUE & UNCLEAR

"Build an app for data."

WHY IT FAILS:

- ✕ Missing context: What kind of data?
- ✕ Missing objective: What is the goal?
- ✕ No constraints or feature requirements
- ⚠ Result: Generic, unusable prototype

Strong Prompt

DETAILED & STRUCTURED

"Create an app where users can enter patient records, review past entries by date, and get alerts when values go above a threshold. Include a dashboard and a detail view."

WHY IT WORKS:

- ✓ **Context:** Patient records
- ✓ **Features:** Entry form, date review, alerts
- ✓ **Views:** Dashboard, detail page

ANATOMY OF A GOOD PROMPT

A	B	C	D
Context	**Objective**	**Features**	**Constraints**
The problem or scenario	What to build	Required functions	Limits & Rules

Key Message: The better the prompt, the better the starting result.

Why Iteration Matters: Draft → Refine → Solution

The first output is rarely the final product. Vibe coding relies on refining drafts into working solutions.

1 Visibility First

The AI generates an initial version ("draft") from your prompt. It allows you to see something concrete immediately, rather than waiting for perfection.

What to expect:
- Basic layout
- Missing features
- "Rough" logic

2 Iterative Dialogue

You "talk" to the draft to improve it. This conversation is where the real building happens, fixing errors and adding detail.

Common Refinements:
- "Add a search bar"
- "Show results as a table"
- "Fix the calculation error"

3 Working Solution

After cycles of feedback, the draft evolves into a usable application that meets your specific needs.

The Outcome:
- Validated logic
- Clean interface
- Ready for use

What Vibe Coding Enables

Democratize Creation

Domain experts can prototype ideas directly without always waiting for technical teams.

Rapid Prototyping

Move from abstract idea to working prototype in minutes rather than weeks.

Lower Barriers

Reduces the deep programming expertise required to build initial solutions.

Better Communication

Bridges the gap between business needs and technical implementation.

Faster Validation

Experiment with concepts and fail fast to find the right solution sooner.

"Vibe coding democratizes creation, but responsible **review and validation** still matter."

SUMMARY

Section Recap

Prompt-First.
Rapid Iteration.

Conversational Creation

Vibe coding is building digital solutions by describing outcomes in **natural language**. The AI generates a starting point which you refine iteratively.

The 4-Step Cycle

The process follows a repeatable flow: **Describe** → **Generate** → **Test** → **Refine**. This replaces the traditional plan-then-code waterfall.

Prompt Quality & Iteration

Strong prompts (Context + Objective + Features + Constraints) and rapid iteration are essential. The first output is just a **starting draft**.

"Vibe coding speeds up the path to prototype."

COMING UP

Designing AI Automation Workflows

Now that we understand the concept of vibe coding, the next step is to learn how to design AI automation solutions in a **practical** and **structured** way.

PREVIEW: THE DESIGN PROCESS

Identify Problem

Find the right automation opportunity

→

Break Down

Map workflow steps & logic

→

Define Output

Clarify results & deliverables

Proceed to Next Section →